

MATH9102: Fundamentals of Data Analysis

Week 2: Describing Data

Distributions, Standardisation and Visualisation

Today

1. **From a Sample to a Population** — and the standard error
2. **The Central Limit Theorem** — stated carefully
3. **The Normal Distribution**
4. **Standardising** — z-scores
5. **Shape** — skew, kurtosis, outliers
6. **Choosing a Picture**
7. **Describing a Whole Dataset**

Where this is going

Everything here is groundwork for week 3. A statistical test compares something you observed against a distribution and asks how surprising it is. This week builds the distribution.

Where We Are

Last week

- A variable's level of measurement is your judgement
- Describe a categorical variable with counts and percentages
- Describe a continuous one with a **centre** and a **spread**
- Look at missingness before dropping anything

This week

- **Shape** — the third thing, and the one that decides which centre to use
- **A picture** — which one, and what it can mislead about
- **Sampling** — how much a statistic would have moved if you had a different sample

The one-sentence version

You have a sample. Everything you compute from it would have come out differently with a different sample, and this week is about how much differently.

From a Sample to a Population

Parameter and Statistic

You want the mean length of every fish in the lake.

Parameter

- A fixed fact about the **population**
- There is exactly one value of it at any moment
- It is almost never known, and usually cannot be

Statistic

- A value computed from a **sample**
- Your estimate of the parameter
- There are as many values of it as there are possible samples

The notation, so it does not surprise you later

Greek letters for parameters (μ for a population mean, σ for a population standard deviation), Latin for the statistics that estimate them ($\bar{x}$, s).

Everyone in this room catches six fish from the same lake.

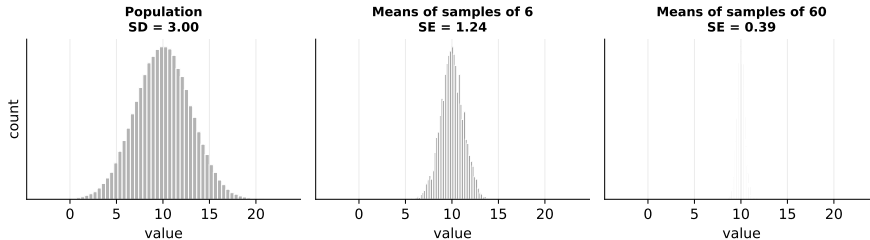
- Would you all catch the *same* fish? No
- Would your mean lengths agree? No
- Would your *ranges* agree? Also no

None of you has made a mistake. This is **sampling variation**, and it is a property of sampling, not of carelessness.

So what is your one sample worth?

That is the question the rest of this section answers, and it is the question every statistical test is ultimately built on.

The Sampling Distribution



Take many samples, compute the mean of each, plot those means. The result is the **sampling distribution of the mean** — a distribution of a statistic, not of the data.

Two things to read off

It is centred on the population mean. And it is far narrower than the population — narrower still as the sample grows.

Standard Error

The standard error is the standard deviation of the sampling distribution.

$$SE = \frac{s}{\sqrt{n}}$$

Large SE

- Sample means vary a lot from sample to sample
- Yours may not represent the population well

Small SE

- Most samples would give much the same mean
- Yours is likely to be close

Watch out

SD and SE are not the same thing. The SD describes how spread out the *observations* are; the SE describes how precisely you have pinned down the *mean*. A large sample of highly variable things has a large SD and a small SE.

Why $\sqrt{n}$, and Not n

If it were n , precision would improve in direct proportion to effort: double the sample, halve the error.

It is $\sqrt{n}$. To halve the standard error you need **four times** the data. To divide it by ten, a hundred times.

Why

Random errors partly cancel each other as you average more of them, and the rate at which they cancel is the square root law. Each extra observation adds less than the one before it.

The practical consequence

Diminishing returns are built into data collection. Past a certain point, spending on a better instrument or a less biased sampling frame buys more than spending on more rows — and no amount of extra rows buys anything at all against *bias*.

Estimating the Standard Error from One Sample

```
# The standard error is the SD of the sampling distribution of the mean.  
# From one sample you can compute it, or bootstrap it: resample your sample  
# with replacement, many times, and see how much the mean moves.  
rng = np.random.default_rng(0)  
sample = m9.load_festival(with_outlier=False).day1.dropna()  
  
se_formula = sample.std(ddof=1) / np.sqrt(len(sample))  
resampled = [rng.choice(sample, len(sample), replace=True).mean() for _ in range(2000)]  
  
print(f"SE from the formula = {se_formula:.4f}")  
print(f"SD of 2000 bootstrap means = {np.std(resampled, ddof=1):.4f}")
```

Two routes, one answer

The second route is the **bootstrap**: resample your one sample with replacement and see how much the mean moves. It agrees with the formula. For a mean, use the formula; the bootstrap earns its keep for statistics that have none.

The Central Limit Theorem

The Central Limit Theorem

The statement

For a sufficiently large sample size, the **sampling distribution of the mean** is approximately normal — **whatever the shape of the population**.

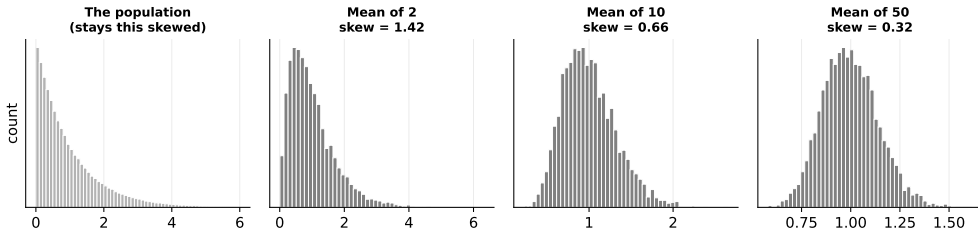
Why this is remarkable

- It does not care what the population looks like
- So you can reason about a sample mean without knowing the population's distribution
- Which is just as well, because you never do

Read the statement again

Every word of it is doing work. Two of them are the ones people drop.

What It Says, and What It Does Not



It does NOT say your data become normal

The population on the left is exactly as skewed as it always was. Nothing you do to a sample changes the population.

It does NOT say a fixed n is enough

At a sample of two the sampling distribution is still visibly skewed. How large is large enough depends on the population.

How Large Is “Large Enough”?

There is no threshold, and you will be told there is.

- For a population that is already close to symmetric, a handful of observations is plenty
- For a strongly skewed one, the sampling distribution can still be visibly skewed at a sample of fifty
- For a heavy-tailed one it can be worse again

Watch out

You will meet claims of the form “at $n = 30$ ” or “at $n = 500$ the CLT is essentially guaranteed, even for strongly non-normal populations”.

They are rules of thumb dressed as theorems. The honest version is: **look at the data, and say what you saw.**

Why the CLT Matters Here

Every test in weeks 3 to 5 does the same thing:

1. Compute a statistic from your sample
2. Work out what that statistic's distribution would look like if nothing were going on
3. Ask how far into the tail of that distribution your value falls

Step 2 is impossible without knowing the shape of the sampling distribution. The CLT is what supplies it.

And when it does not apply

Small samples, badly skewed outcomes, heavy tails — that is precisely when we reach for a rank-based test instead. Week 4.

The Normal Distribution

The Normal Distribution

Its properties

- Symmetric about the mean, and bell-shaped
- Mean, median and mode are the same value
- Approaches the axis but never touches it
- The total area under it is 1

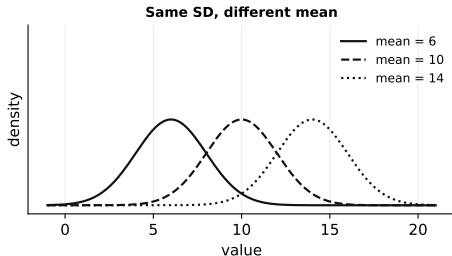
The one that surprises people

It is **completely specified by two numbers**: the mean and the standard deviation. Give me those and I can draw the entire curve. There is nothing else about it that can vary.

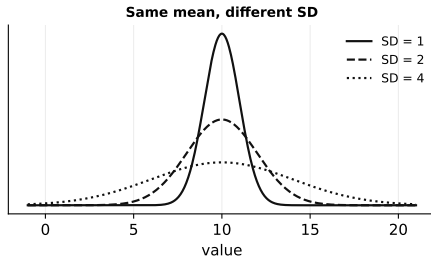
Names

Also called the Gaussian distribution, after Gauss; the bell curve; the normal curve. De Moivre got there first. All the same thing.

Two Numbers Fix It Completely

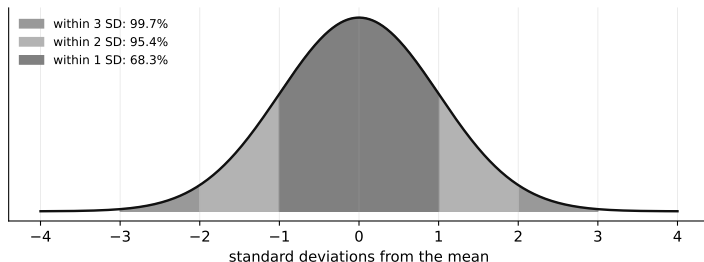


The mean slides the curve along the axis and changes nothing else about it.



The standard deviation sets how wide and how flat it is. A large SD spreads the probability out; a small one concentrates it.

The Empirical Rule



The 68–95–99.7 rule. Worth knowing by heart, because it lets you sanity-check a mean and an SD in your head.

And two more, for later

95% of values lie within 1.96 SDs, and 99% within 2.58. Those two numbers reappear constantly from week 3 onwards.

Density Curves, and Why Area Is Probability

A density curve describes the overall pattern of a distribution — a smoothed histogram with a total area of one.

- The height at a point is *not* a probability
- The **area** between two points is the probability of landing between them
- So the probability of any single exact value is zero — there is no width to it

A uniform distribution makes it obvious

Spread probability evenly from 0 to 50 and the curve is a rectangle of height $1/50$. The probability of landing below 40 is the area of the rectangle from 0 to 40: $40 \times 0.02 = 0.8$.

This is why “the probability that $x < a$ ” and “the probability that $x \leq a$ ” are the same for a continuous variable.

Real Data Is Never Exactly Normal

The normal distribution is a model. Nothing you measure is drawn from one.

- Many things are close enough for the model to be useful
- Being close enough is a judgement, and it is yours
- The question is never “is this normal?” — it is “is it close enough for what I am about to do?”

Why it matters which

Parametric tests assume an approximately normal distribution. **Non-parametric** tests do not.

Choosing between them is a week 4 problem, and it is decided by looking at the data.

Standardising

Two Tests, One Score

Fred sits two tests and scores 78 in both.

	Test A	Test B
Fred's mark	78	78
Class mean	70	66
Class SD	8	6

The raw marks cannot say

They are the same number on two scales with different centres and different spreads. Comparing them compares nothing.

Did he do better in one than the other?

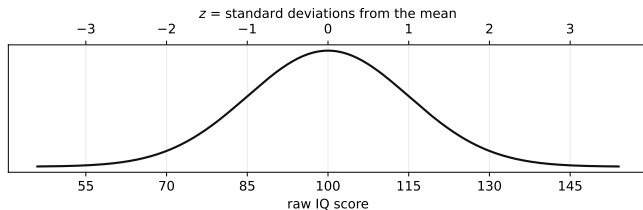
Standardised

$$z_A = (78 - 70)/8 = 1.00 \quad z_B = (78 - 66)/6 = 2.00$$

Fred did considerably better on Test B, relative to the class.

The z-Score

$$z = \frac{x - \bar{x}}{s}$$



One distribution, two rulers. The shape does not change; only the units do.

The set of z-scores always has **mean 0 and SD 1**, whatever the original units were. That is what makes two scales comparable.

The Standard Normal, and the Tables You Do Not Need

A normal distribution with mean 0 and standard deviation 1.

Any normal distribution converts to it by standardising. That is why one table of areas served for every normal distribution — and why generations of textbooks print one.

You have the function instead. It is exact, it takes any value, and it cannot be misread off a row.

What the table gave you

For any point, the area to its left and the area beyond it. The area beyond 2.33 is 0.01: a value that far out happens one time in a hundred.

Two benefits, both still current

Standardising lets you **compare measurements on different scales**, and it turns a raw value into a **probability**.

A Worked Question

```
# A z-score says how many standard deviations a value sits from the mean.  
# Uninjured people average 92 with an SD of 6. Someone with a head injury scores 89.  
z = (89 - 92) / 6  
  
# The area below z is a tail probability. No printed table required.  
print(f"z = {z:.2f}")  
print(f"share of uninjured people this low or lower: {stats.norm.cdf(z):.4f}")  
  
# The same idea on real data: respondents more than 2 SDs above the mean.  
z_stress = (survey.tpstress - survey.tpstress.mean()) / survey.tpstress.std(ddof=1)  
print(f"respondents above z = 2: {(z_stress > 2).sum()}")
```

The classic form. A comprehension test on uninjured people averages 92 with an SD of 6. Someone with a head injury scores 89. Impaired?

Reading the output

Roughly a third of uninjured people would score this low or lower. On this evidence, no.

Why This Matters Later

You have just done, by hand, what every test in this module does.

1. Take an observed value
2. Express it as a distance from what you would expect, in units of the expected variability
3. Ask how far into the tail that puts it

The three facts it rests on

- Scores are distributed around their mean
- Sample means are distributed around the population mean
- Differences between sample means are distributed around zero

The third one is the whole of week 4. A t -statistic is a standardised difference between two means, and the p -value is a tail area — exactly as here.

Shape

Two variables can share a centre and a spread and look nothing alike.

What to describe, for a continuous variable

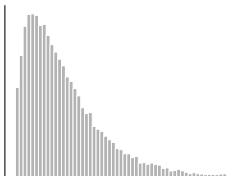
- **Centre** — mean, median, mode
- **Spread** — range, SD, IQR
- **Shape** — symmetric, skewed, bimodal, uniform
- **Unusual features** — outliers, gaps, clusters

And the fifth thing

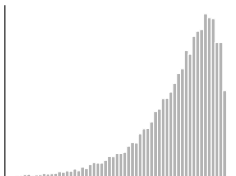
Context. Write it in sentences, about the variable, using its real-world units. A table of numbers with no sentences around it is not a description.

Skew and Kurtosis

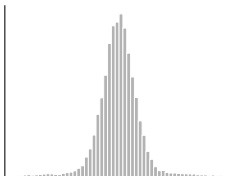
Positive skew
tail to the right
skew 1.60, kurtosis 3.81



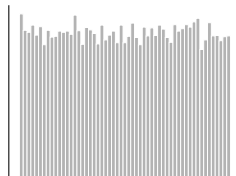
Negative skew
tail to the left
skew -1.58, kurtosis 3.66



Heavy tails
(leptokurtic)
skew 0.01, kurtosis 9.18



Light tails
(platykurtic)
skew -0.00, kurtosis -1.21



Skew is asymmetry. Positive means a tail to the *right* — which is the direction that trips everybody up.

Kurtosis is about the tails: how much of the variance comes from rare extreme values. Reported as *excess*, so zero means “like a normal”.

Reporting Shape

```
# Shape is the third thing to report, after centre and spread.  
m9.describe(festival, ["day1", "day2", "day3"])  
  
# Positive skew means a tail to the right. Kurtosis above zero means more of  
# the variance comes from rare extreme values than a normal would give.  
# Both are descriptions. Neither is a pass/fail test.
```

Watch out

Both numbers are **descriptions**, not tests. You will meet material that converts them to standardised ratios and treats crossing a threshold as a verdict. At a few hundred observations that threshold is crossed by departures far too small to matter — week 3 shows why.

Outliers, and the $1.5 \times \text{IQR}$ Fence

The conventional screen: more than $1.5 \times$ the IQR beyond a quartile.

- Upper fence: $Q3 + 1.5 \times \text{IQR}$
- Lower fence: $Q1 - 1.5 \times \text{IQR}$
- Values beyond a fence are drawn individually on a boxplot

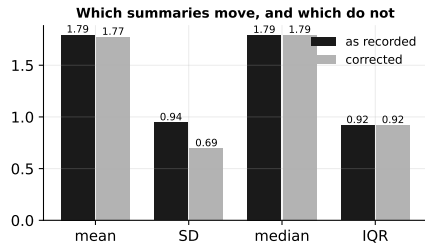
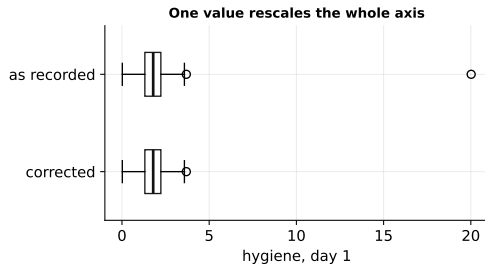
A screen, not a verdict

On a **skewed** variable the rule flags ordinary observations in the tail as a matter of arithmetic. Nothing is wrong with them, and deleting them makes the sample less representative.

What to do with a flagged value

Find out what it is. If it is impossible given the scale, it is an error: correct it, or exclude it and say so. If it is genuine, it stays — report the skew and use summaries that survive it.

What One Bad Value Does



Festival hygiene is rated from 0 to 4. One record says twenty. That is a typing error, not a very clean person.

Resistant summaries

The mean and SD move. The median and IQR do not notice.

Finding Them

```
# The conventional fence: 1.5 x IQR beyond the quartiles.
q1, q3 = festival.day1.quantile([0.25, 0.75])
iqr = q3 - q1
fence = q3 + 1.5 * iqr

flagged = festival.loc[festival.day1 > fence, "day1"]
print(f"upper fence = {fence:.2f}, values beyond it: {len(flagged)}")
print(f"largest: {flagged.max():.2f}")

# Being beyond the fence is not a reason to delete a value. It is a reason to
# find out what it is.
```

Two habits

Check the flagged values against what the variable *can* be — that is what turns a flag into a decision. And report how many you found and what you did, every time.

Choosing a Picture

The Art of Presenting Data

A graph should (Tufte, 2001):

- Show the data
- Make the reader think about the *substance*, not about the graph
- Avoid distorting what the data have to say
- Present many numbers in a small space
- Make a large dataset coherent
- Encourage the eye to compare
- Reveal the data at several levels of detail
- Serve a clear purpose: description, exploration, tabulation or decoration
- Sit closely with the statistical and verbal description

The test that catches most bad charts

Could a reader reconstruct roughly what the numbers were from your picture? If not, the picture is decoration.

Which Plot for Which Variable

What you have	Plot	Summary statistics
One continuous variable, roughly symmetric	histogram or density plot	mean and SD
One continuous variable, skewed	histogram <i>and</i> boxplot	median and IQR
Continuous, compared across groups	boxplots side by side	per group, as above
One categorical variable	bar chart, or a frequency table	counts, percentages, mode
Two continuous variables	scatterplot	correlation (week 3)

The rule underneath the table

The plot follows the **level of measurement**, and the summary statistic follows the **shape**. Decide the first from the variable and the second from the picture.

Histograms

What it is

- The horizontal axis carries the values of one variable
- The height of each bar is how many observations fall in that interval
- Or the *relative* frequency, which makes different-sized datasets comparable

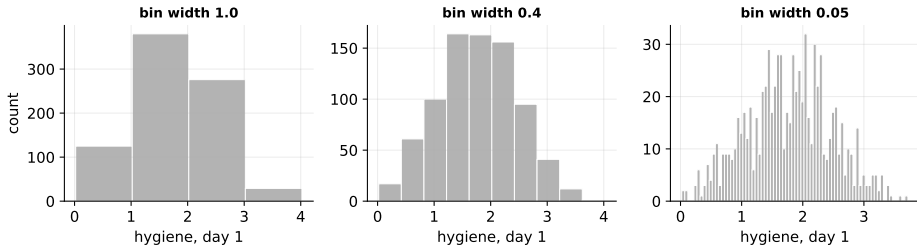
How to read one

- Where is the centre?
- How wide is the spread?
- Is it symmetric, skewed, or does it have more than one peak?
- Anything isolated out in a tail?

Discrete or continuous

For a **discrete** variable put one bar over each possible value. For a **continuous** one you have to choose intervals — and that choice is not innocent.

The Bin Width Is Your Choice

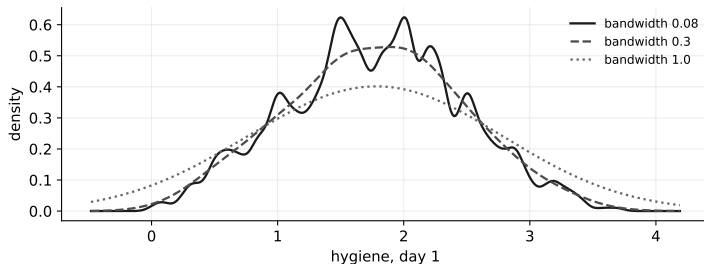


One dataset, three histograms. Too wide and the shape disappears; too narrow and you are reading noise.

Watch out

A histogram is a fact about the data *and* your bin width. Try several before you believe any of them.

Density Plots, and Bandwidth



A density plot is a smoothed histogram. Each observation contributes a small bump; the bumps are added up.

The **bandwidth** controls how wide each bump is, and does exactly what the bin width does. Same warning.

Stem-and-Leaf

A histogram that keeps the actual values.

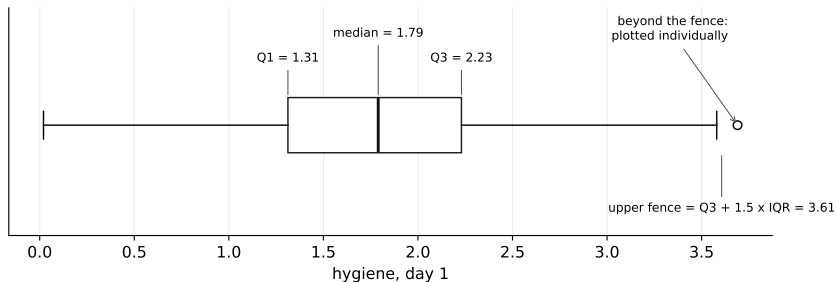
Stem	Leaves
7	5 9
8	3 4 6 6 8
9	1 4 9

Key: 7 | 5 means 75.

- Split each value into a leading digit (the stem) and a trailing one (the leaf)
- The shape of the plot is the shape of the distribution
- Unlike a histogram, **nothing is lost** — you can read the median and the range straight off it
- Good for small and moderate datasets; useless for large ones

Largely a pen-and-paper technique now, and worth ten minutes because it makes clear what a histogram is doing.

Boxplots: The Five-Number Summary, Drawn



The box spans the middle half of the data, so its height *is* the IQR. The whiskers reach the most extreme values still inside the fences.

The fences are scaffolding

They decide where the whiskers stop and which points are drawn individually. They are not part of the finished plot.

Reading a Boxplot

Read it in this order.

1. **Where the median sits in the box.**

Off-centre means the middle half is skewed

2. **How tall the box is.** That is the IQR

3. **Whether the whiskers match.**

Unequal whiskers mean skew in the tails

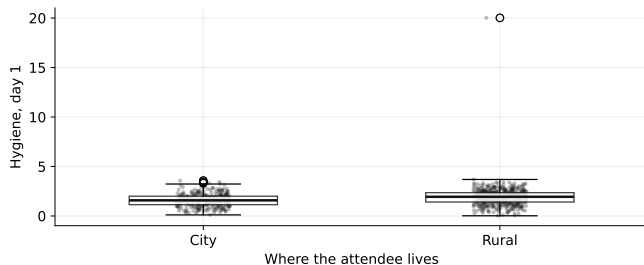
4. **What is beyond them**

What it hides

A boxplot cannot show you a **bimodal** distribution. Two clear peaks and one smooth hump can produce identical boxplots.

That is the trade: it hides the detail so that several groups can be compared at once. Look at a histogram too.

Comparing Groups

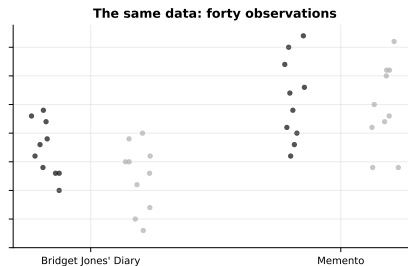
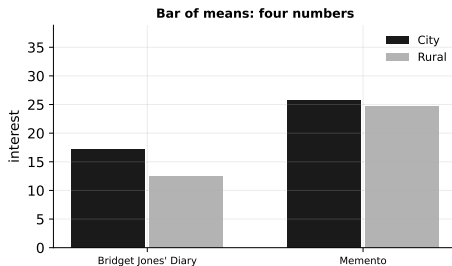


Boxplots side by side are the best value-for-ink comparison there is: centre, spread, skew and outliers for every group at once.

Watch out

Still a description. Overlapping boxes are not a test, and non-overlapping ones are not a test either.

Bar Charts, and One to Avoid



A bar chart of **counts** is the right plot for a categorical variable. A bar chart of **group means** is not.

Same data, same ink

The left panel shows four numbers. The right, a dot plot, shows forty observations, their spread, and their overlap.

Plots in Python

```
# One plot per question. The bin width is a choice you are making, so make it
# deliberately and try more than one.
m9.histogram_with_normal(festival.day1, xlabel="Hygiene, day 1")

# Comparing groups: a boxplot shows centre, spread, skew and outliers at once.
m9.grouped_box(festival, "day1", "location", ylabel="Hygiene, day 1")
```

The module's helpers cover the plots this course needs and set a consistent style. Underneath they are matplotlib, and you can reach past them whenever you want something else.

A reminder

This is not a module in visualisation. One correct plot per question, labelled, beats a gallery.

Describing a Whole Dataset

What Kind of Study Is It?

Say how the data came to exist. It decides what they can answer.

Design	How the data arise	Example
Observational	recorded as it happens; nothing is changed	library hours against grades
Survey	structured questions elicit the data	this week's wellbeing questionnaire
Experiment	the researcher changes a condition, ideally at random	music against silence, then a stress scale
Meta-analysis	earlier studies' results are combined	pooling published effects

“Did it work?” needs a comparison the researcher assigned

Observational and survey data describe and show association; they cannot rule out that something else produced the difference. A meta-analysis inherits the designs it pools.

The Case Study

A survey of psychological adjustment and wellbeing among members of the general public in Melbourne, Australia.

- Validated scales for self-esteem, optimism, perceived control, perceived stress, positive and negative affect, and life satisfaction
- Plus a social-desirability scale — a measure of the tendency to answer flatteringly
- Plus demographics

Read that last one again

A scale measuring how much respondents present themselves favourably is a measurement of a *bias in the other measurements*. Somebody designed this study carefully.

Participation was voluntary, so the sample is self-selected. That belongs in the description, not in a footnote.

Type Every Variable You Will Use

Variable	Concept	Level
sex	Gender	nominal
age	Age in years	ratio
marital	Marital status, eight categories	nominal
educ	Highest level of education completed	ordinal
smoke	Smoker	nominal
tpstress	Perceived Stress Scale total	interval, by convention

Watch out

educ is the one people get wrong. Its categories are ranked, so it is ordinal — but they are stored as text, so software sorts them alphabetically and “primary” lands after “postgraduate”. The order is yours to impose.

Many Items, One Measure

Each psychometric scale is a set of Likert items. Analysing them individually would be unmanageable and beside the point.

So the dataset carries a **total** for each construct: toptim, tpstress, tlifesat, tslfest, tposaff, tnegaff, tpcoiss, tmast, tmarlow.

Establishing that a set of items *can* be combined this way is **dimension reduction**, in week 10.

The honest position on their level

Adding up ordinal items does not, strictly, produce an interval measure.

Validated scales are treated as continuous throughout the literature and throughout this module. That is a **defensible convention**, not a mathematical fact, and you should be able to say which it is.

What Your Reader Needs

Enough that they could reproduce the study, and enough that they can see what constrains your conclusions.

1. **Who** the participants were, and how they were identified
2. **How** they were sampled — invited, self-selected, incentivised? Secondary data?
3. **What** was measured, with what instrument, and what is known about its validity and reliability
4. **Each variable** you will use: level of measurement, possible values, summary statistics
5. **What is missing**, and what it costs
6. **Sources**, for the data and for the methodological choices

Only the variables your question needs

This dataset has 134 columns. A description of all of them is not a description; it is a data dump. Name the question first.

A Worked Description

```
# The description a reader needs, in the order they need it.  
survey = m9.load_survey()  
  
m9.missingness(survey, ["toptim", "tlifesat", "tpstress", "tslfest"])  
m9.frequency(survey, "sex")  
m9.describe(survey, ["age", "toptim", "tlifesat", "tpstress", "tslfest"])
```

Every number in the write-up is generated

The notebook builds the description paragraph from these objects, so it cannot drift from the data. Change the data, run it again, and the paragraph changes with it. **This is the standard for every description you write.**

Which Statistic, Which Plot

Interval or ratio

Roughly normal: mean and standard deviation; histogram.

Skewed or heavy-tailed: median and IQR; histogram *and* boxplot.

Nominal or ordinal

Mode and frequencies; percentages, saying which way you percentaged; bar chart or table.

Ordinal additionally permits a median.

Provide the maximum description with the minimum statistics

Not every statistic for every variable. The right ones for the variables your question needs, each with a sentence saying what it tells the reader.

Close

What you should take from this week

You SHOULD be able to

- Distinguish a parameter from a statistic, and an SD from an SE
- Explain sampling variation, and what a sampling distribution is
- State the Central Limit Theorem accurately, and say what it does not claim
- Use the empirical rule to sanity-check a mean and a standard deviation
- Convert a value to a z-score and turn it into a tail probability
- Describe a continuous variable's centre, spread *and* shape, and choose the summaries that suit its shape
- Apply the IQR fence, and decide what a flagged value actually means
- Choose the right plot, and say what your bin width or bandwidth is doing to it
- Write a dataset description a reader could act on

That's all for this week